

An Explanation of Discussion 5

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1 Intro

During discussion today, there was some confusion regarding the correct procedure for solving the question presented in the fifth discussion assignment. Here, I'll give a general, brief outline of the way the problem was intended to be solved.

2 Overview

In the problem, we were asked to approximate the location of the peak of Sugarloaf Mountain. To do so, we had to estimate the critical points of the function

$$s(x, y) = e^{-(x-1)^2-(y-2)^2} + \frac{3}{4}e^{-(x-2)^2-(y-1)^2}.$$

Traditionally, we would find critical points by solving

$$\nabla s(x, y) = \left\langle \frac{\partial s}{\partial x}(x, y), \frac{\partial s}{\partial y}(x, y) \right\rangle = \langle 0, 0 \rangle.$$

However, when we calculate $\nabla s(x, y)$, both $\frac{\partial s}{\partial x}(x, y)$ and $\frac{\partial s}{\partial y}(x, y)$ are generally sort of 'messy' equations, and solving them for zero may be difficult. So, to deal with this, we will instead *approximate* when they are zero using an iterative method.

How we're supposed to go about this is by applying Newton's Method to the gradient of $s(x, y)$, e.g. $\nabla s(x, y)$. That is, starting with a point (x_0, y_0) , we want to solve

$$\begin{aligned} \frac{\partial f_1}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f_1}{\partial y}(x_0, y_0)(y - y_0) &= -f_1(x_0, y_0) \\ \frac{\partial f_2}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f_2}{\partial y}(x_0, y_0)(y - y_0) &= -f_2(x_0, y_0) \end{aligned}$$

where we use f_1 and f_2 as convenient shorthand to denote

$$f_1(x, y) = \frac{\partial s}{\partial x}(x, y)$$

$$f_2(x, y) = \frac{\partial s}{\partial y}(x, y).$$

We will call the solution to this (x_1, y_1) . After finding this solution, we then repeat the same process again, replacing (x_0, y_0) in the system above with (x_1, y_1) to yield a solution (x_2, y_2) and so on.

Now, in the statement of the problem, we set $(x_0, y_0) = (1, 2)$. So, to use this and find our solution (x_1, y_1) , we first need to calculate the partial derivatives of f_1 and f_2 . After we've done that, using $(x_0, y_0) = (1, 2)$ and the partial derivatives calculated above, our system

$$\begin{aligned}\frac{\partial f_1}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f_1}{\partial y}(x_0, y_0)(y - y_0) &= -f_1(x_0, y_0) \\ \frac{\partial f_2}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f_2}{\partial y}(x_0, y_0)(y - y_0) &= -f_2(x_0, y_0)\end{aligned}$$

becomes

$$\begin{aligned}\frac{\partial f_1}{\partial x}(1, 2)(x - 1) + \frac{\partial f_1}{\partial y}(1, 2)(y - 2) &= -f_1(1, 2) \\ \frac{\partial f_2}{\partial x}(1, 2)(x - 1) + \frac{\partial f_2}{\partial y}(1, 2)(y - 2) &= -f_2(1, 2)\end{aligned}$$

which has a solution

$$\begin{aligned}x &= x_1 \\ y &= y_1,\end{aligned}$$

where x_1 and y_1 are the numbers you get from solving the system above.

So, we then use (x_1, y_1) in place of (x_0, y_0) and repeat the process. That is, we solve

$$\begin{aligned}\frac{\partial f_1}{\partial x}(x_1, y_1)(x - x_1) + \frac{\partial f_1}{\partial y}(x_1, y_1)(y - y_1) &= -f_1(x_1, y_1) \\ \frac{\partial f_2}{\partial x}(x_1, y_1)(x - x_1) + \frac{\partial f_2}{\partial y}(x_1, y_1)(y - y_1) &= -f_2(x_1, y_1)\end{aligned}$$

and produce a solution (x_2, y_2) . We continue to carry out this process (which will minimize the gradient) until we are accurate to within two decimal places of one of the potential answers provided (and thus close to the peak).